Policy gradient methods

Lecture 2: baselines, actor-critic, TD, GAE and PPO

Carlos Cotrini

ETH Zurich

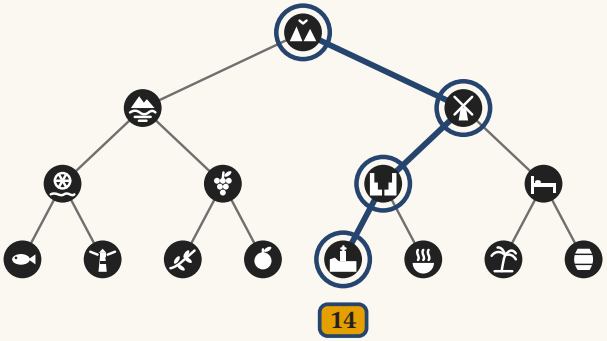
20 October 2026

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Part 5

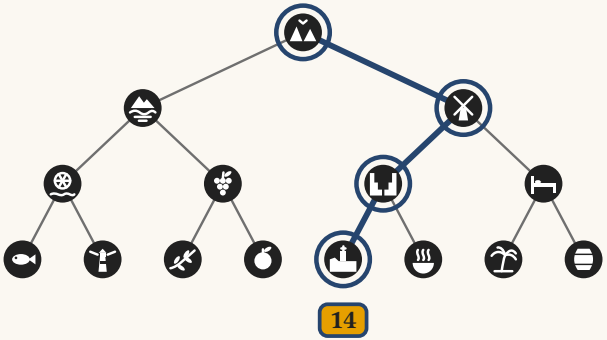
REINFORCE with a baseline

G_t alone is not very informative



1 One day on the outlined path paid $G_1 = 14$ coins, luck included.

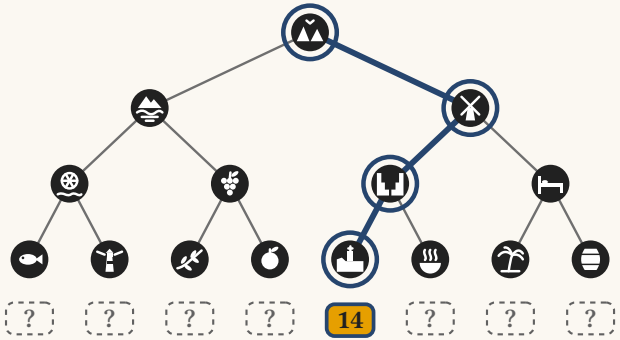
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1 One day on the outlined path paid $G_1 = 14$ coins, luck included.

2 Is this path good? Is it bad?

G_t alone is not very informative

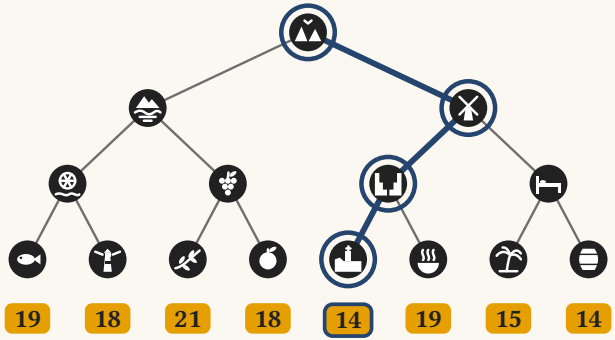


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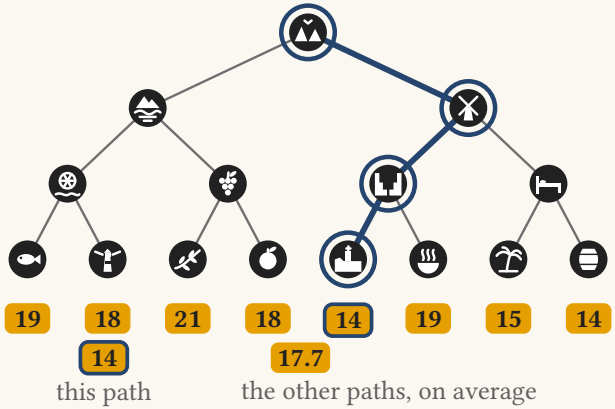
3 It depends on what the other paths yield.

If the other paths pay more, this path is poor



1 The gold numbers are what a day on each path pays on average.

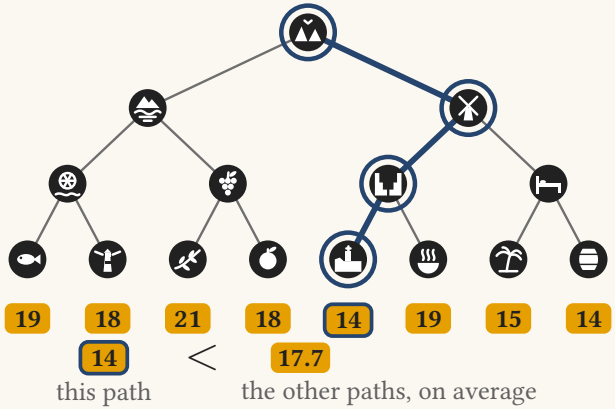
If the other paths pay more, this path is poor



1 The gold numbers are what a day on each path pays on average.

2 The other 7 paths average 17.7 coins, more than the 14 of this day.

If the other paths pay more, this path is poor

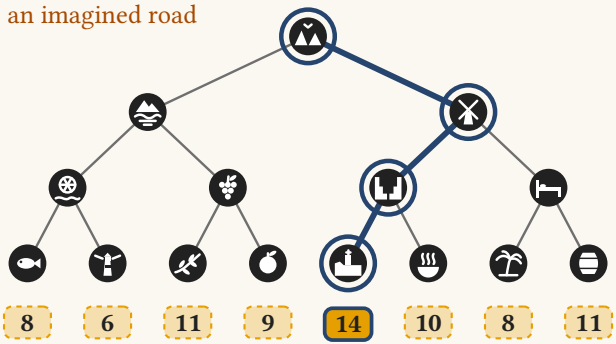


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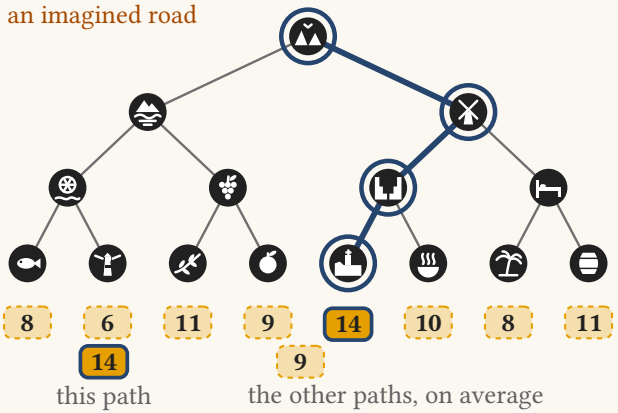
3 Next to them, this path is poor.

If the other paths pay less, this path is good



1 On an imagined road the other paths would pay the dashed numbers.

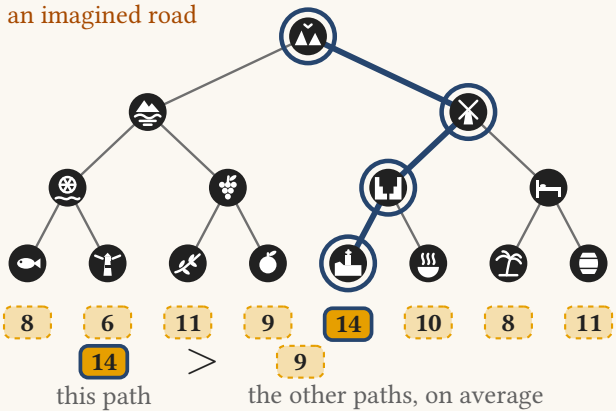
If the other paths pay less, this path is good



1 On an imagined road the other paths would pay the dashed numbers.

2 The other 7 paths average 9 coins, less than the 14 of this day.

If the other paths pay less, this path is good



1 On an imagined road the other paths would pay the dashed numbers.

2 The other 7 paths average 9 coins, less than the 14 of this day.

3 Next to them, the same path is good. The multiplier should include what the other paths yield.

The multiplier compares a path with the average

$$\Psi_t = G_t$$

1 Part 4's multiplier is $\Psi_t = G_t$. Coins alone do not say whether a path is good.

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1. Part 4's multiplier is $\Psi_t = G_t$. Coins alone do not say whether a path is good.
2. The new multiplier subtracts $V(s_t)$. A number subtracted like this is a baseline.

The multiplier compares a path with the average

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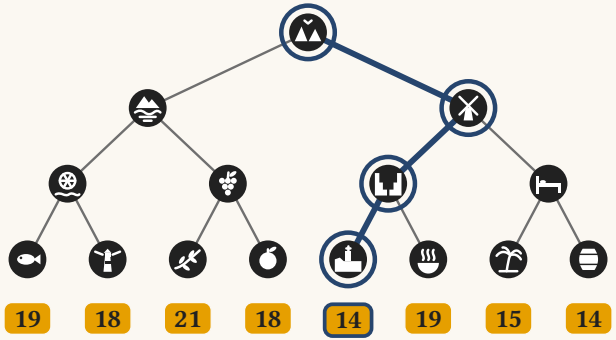
the average of G_t over all the paths from s_t

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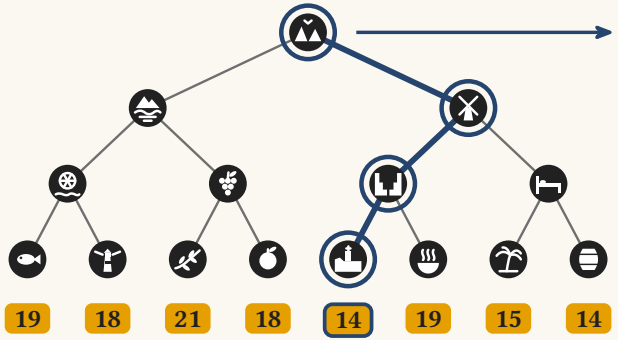
3 $V(s_t)$ averages G_t over all the paths that π_w can take from s_t, each weighted by its chance, the path itself included.

Example: the uniform policy gives $V(s_1) = 17.25$



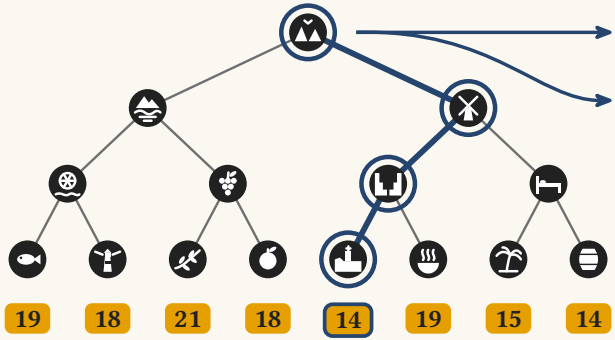
1 The uniform policy: every road has chance 1/2, so each path has chance 1/8.

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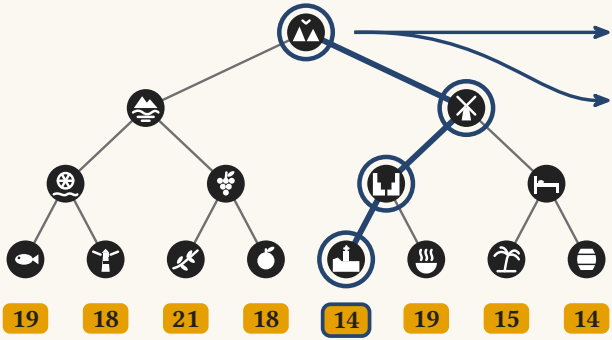


$G_1 = 14$

$V(s_1)$: the average of G_1 over all 8 paths

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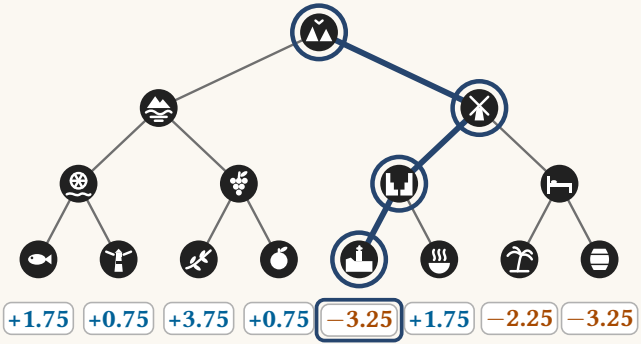
$G_1 = 14$

$V(s_1)$: the average of G_1 over all 8 paths

1. The uniform policy: every road has chance 1/2, so each path has chance 1/8.
2. The outlined path is one of the eight. The other 7 average 17.7.

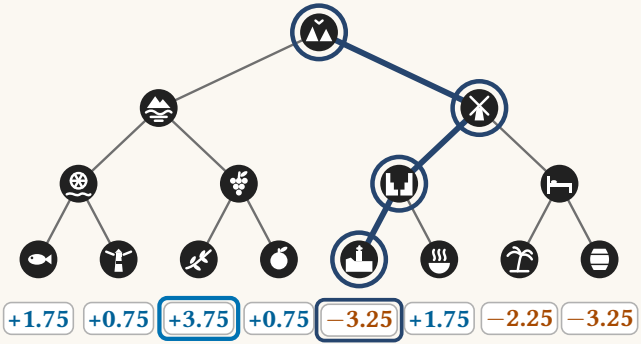
$$V(s_1) = (19 + 18 + 21 + 18 + 14 + 19 + 15 + 14) \,/\, 8 = 17.25$$

The best path gets the largest positive multiplier



1 For each path, $\Psi_1 = G_1 - V(s_1)$. For the outlined path, $14 - 17.25 = -3.25$.

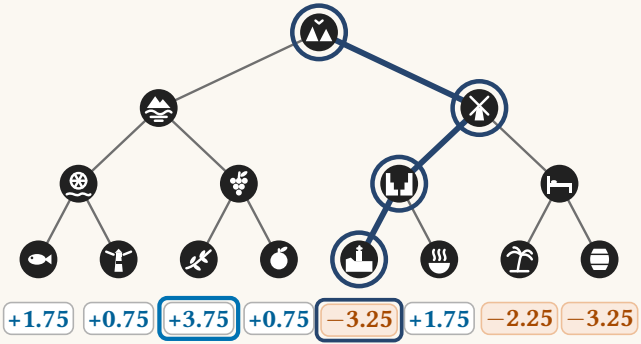
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2 The best path, 21 coins, gets the largest multiplier, +3.75.

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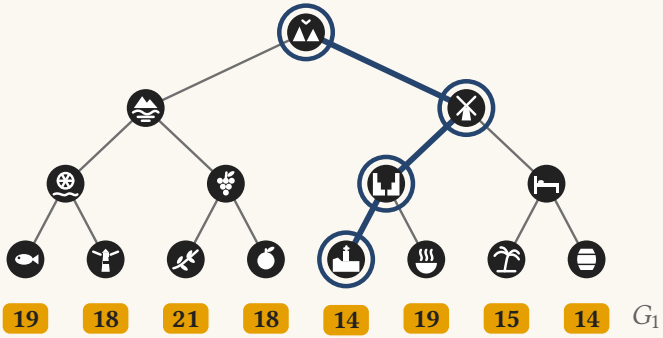


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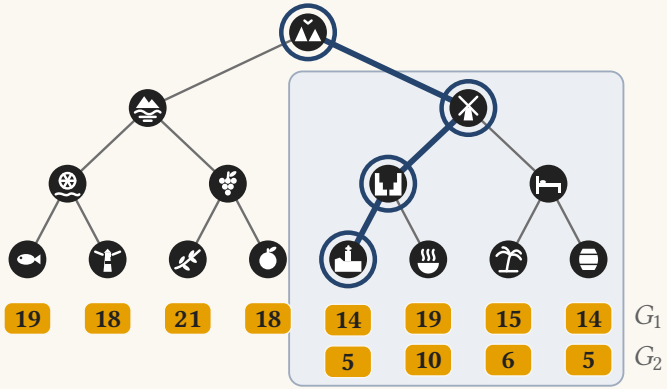
3 The three paths below the average get a negative multiplier.

$V(s_t)$ depends on where Mara stands



1 $V(s_t)$ averages G_t over the paths still ahead. At step 1 all 8 are: $V(s_1) = 17.25$.

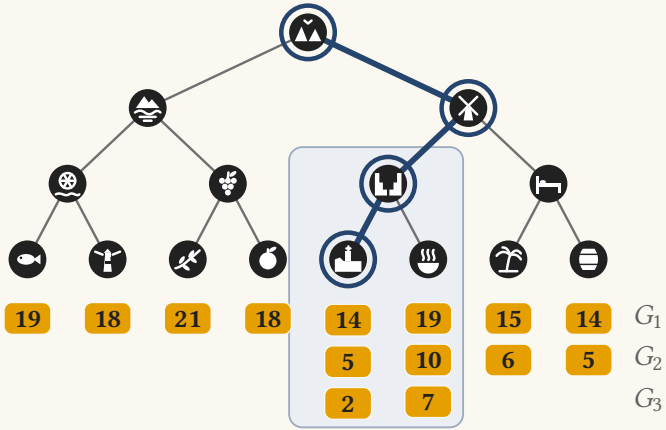
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2 At step 2 Mara stands at the market and 4 paths remain: $V(s_2) = 6.5$.

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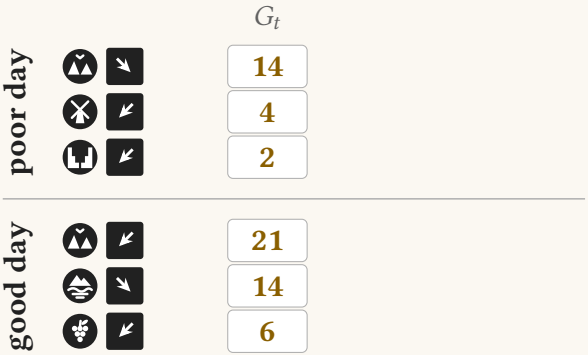


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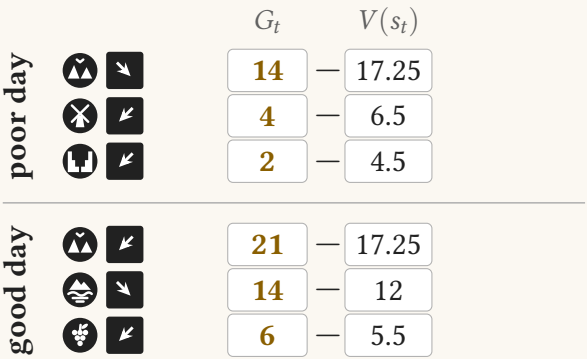
3 At step 3 she stands at the canyon and 2 remain: $V(s_3) = 4.5$.

The poor day now pushes its roads down



1 Part 4 pushed every road by G_t: even the poor day pushes its three roads up.

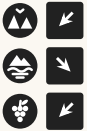
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2 $V(s_t)$ is the average of G_t from the place where the road was taken.

The poor day now pushes its roads down

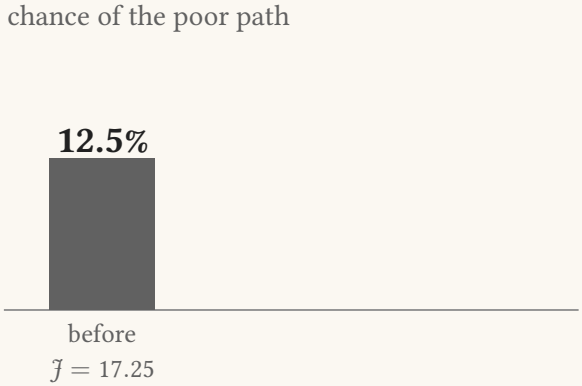
		G_t		$V(s_t)$		Ψ_t
poor day		14	−	17.25	=	−3.25
		4	−	6.5	=	−2.5
		2	−	4.5	=	−2.5
good day		21	−	17.25	=	+3.75
		14	−	12	=	+2
		6	−	5.5	=	+0.5

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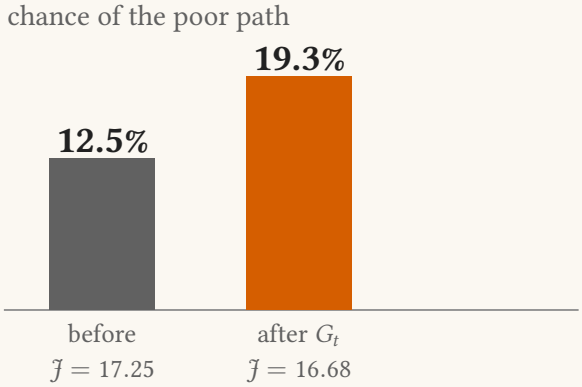
3 The new push is $G_t - V(s_t)$: the poor day's roads go down, the good day's stay up.

With the baseline the poor path becomes rarer



1 From the uniform table, update the knobs with the poor day alone, with step size 0.05.

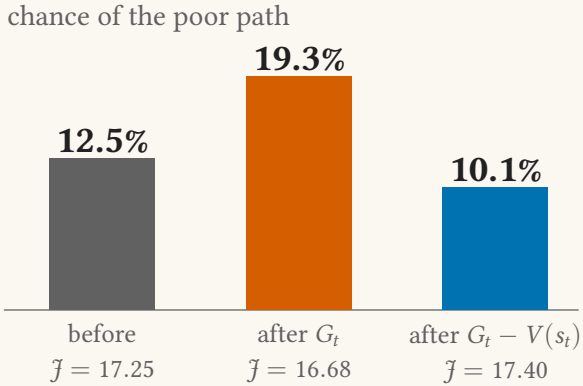
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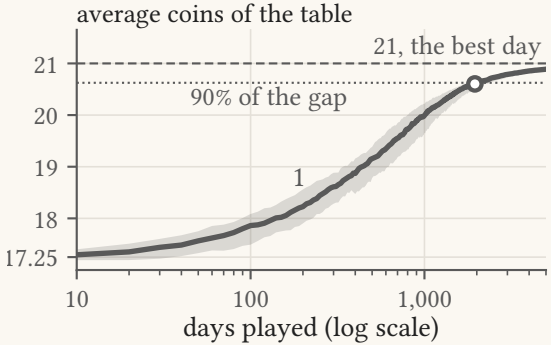


1 From the uniform table, update the knobs with the poor day alone, with step size 0.05.

2 With G_t the poor path becomes likelier and $\mathcal{J}$ falls.

3 With $G_t - V(s_t)$ it becomes rarer and $\mathcal{J}$ rises. One day is a small sample: on average the two updates agree.

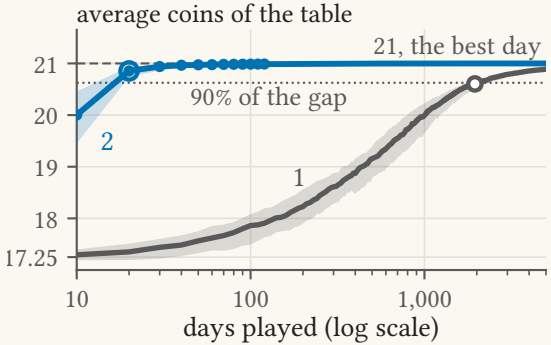
REINFORCE with a baseline converges much faster



Median of 200 runs, with the middle half. The safe step is the fastest at which 99 of 100 other runs close 90% of the gap.

1 REINFORCE at its safe step, 0.0316, closes 90% of the gap from 17.25 to 21 coins in a median of 1,950 days.

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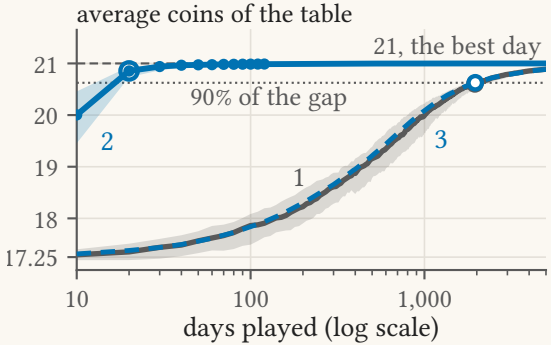


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2 With the baseline, at its safe step, 5.62, it takes 20 days: 97.5 times fewer. Here $V(s_t)$ is exact.

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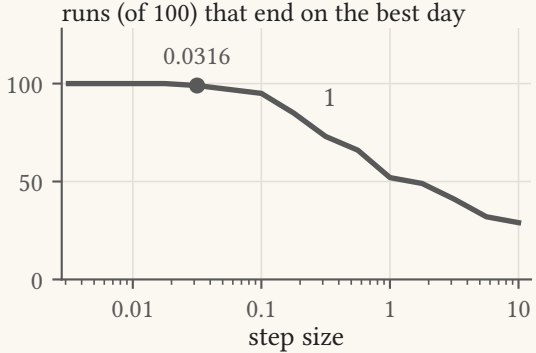
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3 At REINFORCE's step the same baseline is no faster: 1,960 days.

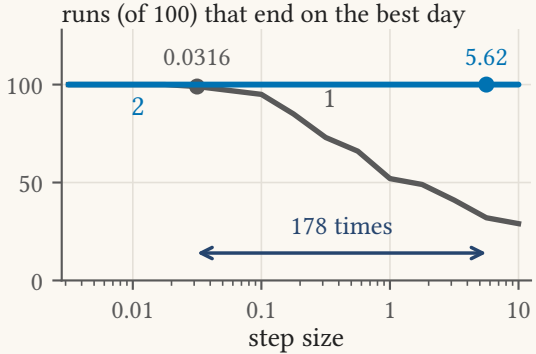
The baseline allows a much larger step



A run ends on the best day when its likelier roads lead to the 21-coin path.

1 REINFORCE fails more often above 0.0316: at 5.62, 32 of 100 runs end on the best day.

The baseline allows a much larger step

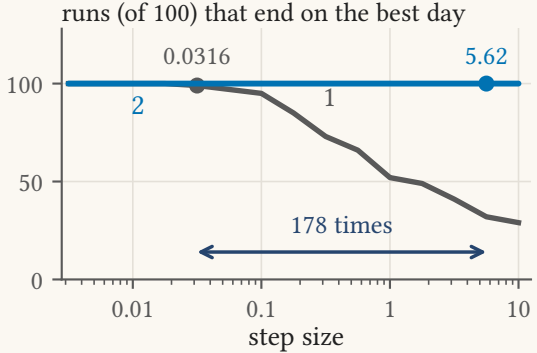


A run ends on the best day when its likelier roads lead to the 21-coin path.

1 REINFORCE fails more often above 0.0316: at 5.62, 32 of 100 runs end on the best day.

2 With the baseline all 100 do, up to 10, the largest step tried. Its safe step is 178 times larger.

The baseline allows a much larger step



A run ends on the best day when its likelier roads lead to the 21-coin path.

1 REINFORCE fails more often above 0.0316: at 5.62, 32 of 100 runs end on the best day.

2 With the baseline all 100 do, up to 10, the largest step tried. Its safe step is 178 times larger.

3 At one common step the two take equally long, since their average push is the same. The baseline's push is quieter, and that allows the larger step.

Here $V(s_t)$ is computed from the road itself

```
w = knobs, all 0
repeat:
  compute V(s) for every place
  play N days with softmax(w)
```

1 $V(s)$ of every place, from the table and the road: the tree is small, so it is exact.

Here $V(s_t)$ is computed from the road itself

```
w = knobs, all 0
repeat:
  compute V(s) for every place
  play N days with softmax(w)
  push every road of every day
    as hard as the coins after it
    minus V(s_t)
```

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2 Every road is pushed by $G_t - V(s_t)$ instead of G_t.

Here $V(s_t)$ is computed from the road itself

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w = knobs, all 0
repeat:
1   compute V(s) for every place
    play N days with softmax(w)
    push every road of every day
2     as hard as the coins after it
      minus V(s_t)
3   w = w + alpha * average push
```

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2 Every road is pushed by $G_t - V(s_t)$ instead of G_t.

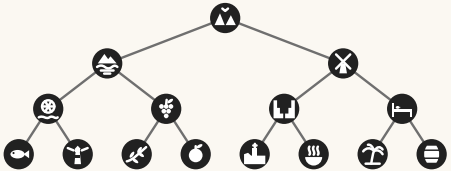
3 The knobs move as before. Every run of the last two frames used this exact V.

6

Part 6

Actor-critic

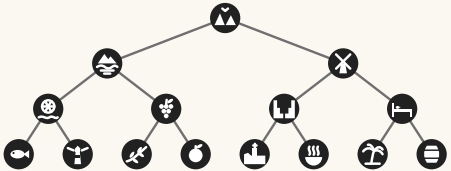
$V(s_t)$ is usually very hard to calculate



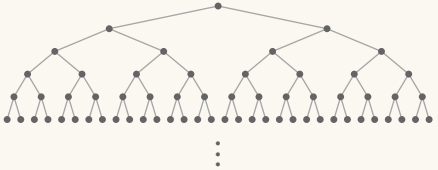
Mara's road: eight paths

1 Part 5's multiplier $\Psi_t = G_t - V(s_t)$ needs $V(s_t)$: on Mara's road an average over eight paths. We listed them.

$V(s_t)$ is usually very hard to calculate



Mara's road: eight paths

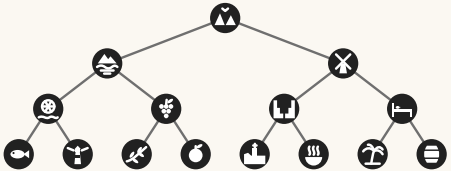


chess, a language model

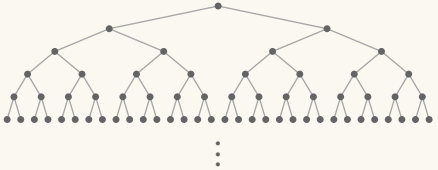
1 Part 5's multiplier $\Psi_t = G_t - V(s_t)$ needs $V(s_t)$: on Mara's road an average over eight paths. We listed them.

2 In chess, or with a language model, no one can list the paths: $V(s_t)$ is very hard to calculate.

$V(s_t)$ is usually very hard to calculate



Mara's road: eight paths



chess, a language model

1 Part 5's multiplier $\Psi_t = G_t - V(s_t)$ needs $V(s_t)$: on Mara's road an average over eight paths. We listed them.

2 In chess, or with a language model, no one can list the paths: $V(s_t)$ is very hard to calculate.

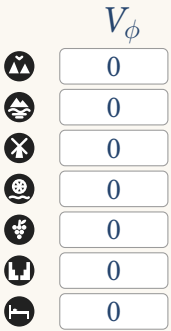
So we estimate it!

the actor
π_w, the table of knobs

→

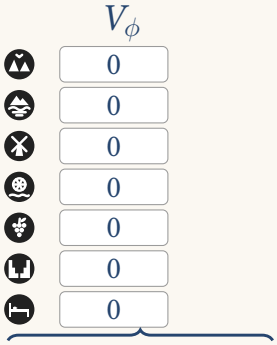
the critic
V_ϕ, an estimate of V

A critic is a function that estimates V



1 A critic is a function V_ϕ: from a place it returns a number, our estimate of V. Its weights are ϕ.

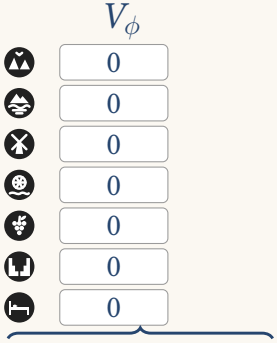
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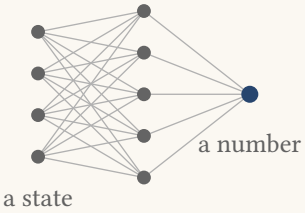
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2 On Mara's road it is a table, one number for each place with a choice. We start it at zeros.

A critic is a function that estimates V



a day that ends is worth 0: no entry



a neural network

1 A critic is a function V_ϕ: from a place it returns a number, our estimate of V. Its weights are ϕ.

2 On Mara's road it is a table, one number for each place with a choice. We start it at zeros.

3 In real life a neural network reads the state. Either way $V_\phi(s_t)$ takes the place of $V(s_t)$ in Ψ_t.

Every step of every day is a labelled example



1 Play N days with π_w: here the five days of Part 2.

Every step of every day is a labelled example

	step 1	step 2	step 3
day 1	15	6	2
day 2	17	10	3
day 3	14	6	1
day 4	15	6	3
day 5	19	9	6

1 Play N days with π_w: here the five days of Part 2.

2 Each step is an example: the place s_t where it starts, and its label, the coins G_t from there to the end of the day.

Every step of every day is a labelled example



1 Play N days with π_w: here the five days of Part 2.

2 Each step is an example: the place s_t where it starts, and its label, the coins G_t from there to the end of the day.

3 It is a data set: learn to guess G_t from the place s_t.

The critic's loss is the squared gap

$$L(\phi) = \frac{1}{N} \sum_{n=1}^{N} \sum_{t=1}^{T} \tfrac{1}{2} \Big(\underbrace{G_t^{(n)} - V_\phi(s_t^{(n)})}_{\text{the gap}} \Big)^2$$

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the labels of each place

- **15, 17, 14, 15, 19**
- **10**
- **6, 6, 6, 9**
- none
- **3**
- **2, 6**
- **1, 3**

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	the labels of each place	their average
	15, 17, 14, 15, 19	**16.0**
	10	**10.0**
	6, 6, 6, 9	**6.75**
	none	none
	3	**3.0**
	2, 6	**4.0**
	1, 3	**2.0**

1 One example costs half the squared gap between the critic's number and its label.

2 Each place has its own part of the loss: its labels only.

3 A part is smallest at the average of its labels: with many days, $V(s_t)$.

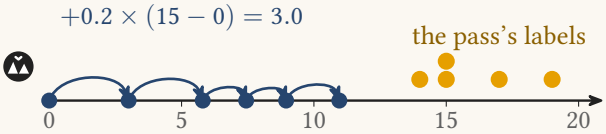
A gradient step moves the table towards the labels

$$\phi \leftarrow \phi - \beta \nabla_\phi \tfrac{1}{2}\big(G_t - V_\phi(s_t)\big)^2$$

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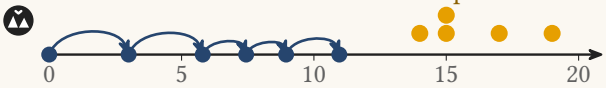


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$+0.2 \times (15 - 0) = 3.0$



	before	after five days
	0	10.96
	0	2.00
	0	4.14
	0	0.00
	0	0.60
	0	1.52
	0	0.76

1 Only the entry of s_t is in the loss of an example: it moves a fraction β of the way to the label. β is the critic's step.

2 One step per example, in order. Here $\beta = 0.2$. A place that no day visited keeps its number.

Algorithm to estimate the critic

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w, phi = zeros, zeros
repeat:
  play N days with softmax(w)
  G = coins to come
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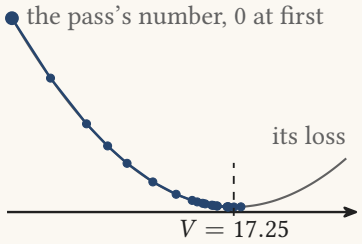
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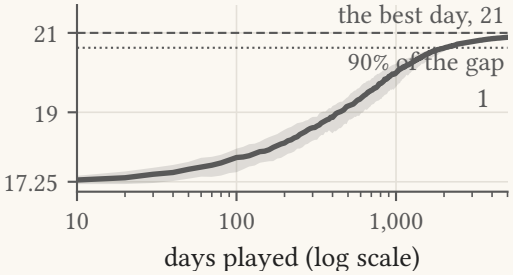
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w, phi = zeros, zeros
repeat:
  play N days with softmax(w)
  G = coins to come
  Psi = G - phi[s]                    1
  w = w + alpha * average push
  for each example, in order:         2
    phi -= beta * grad(loss)
```



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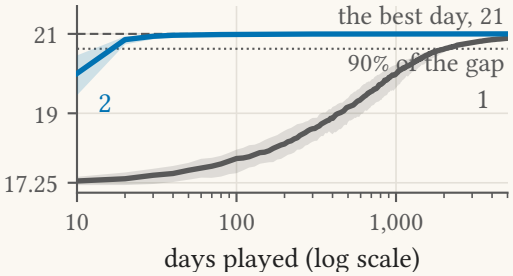
2 The critic steps down its own bowl, one example at a time.

A learned critic speeds REINFORCE up



1 REINFORCE closes 90% of the gap in 1,950 days ($\alpha = 0.0316$; median of 200 runs).

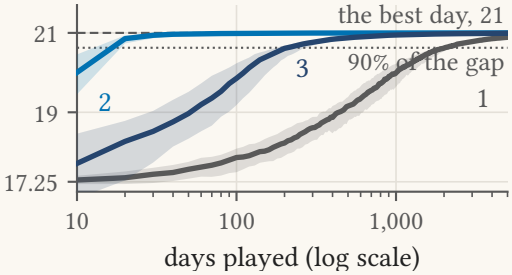
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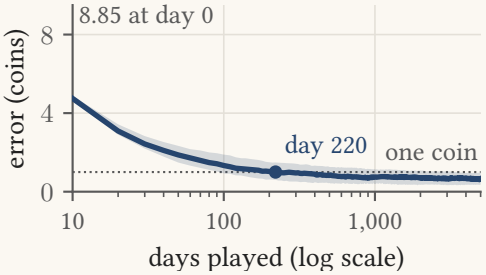


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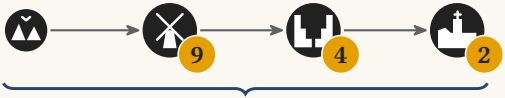
3 A critic learned from zeros: 210 days ($\alpha = 0.316$, $\beta = 0.1$).

7

Part 7

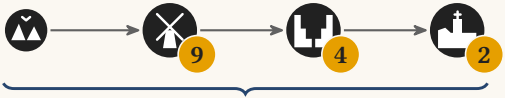
TD learning

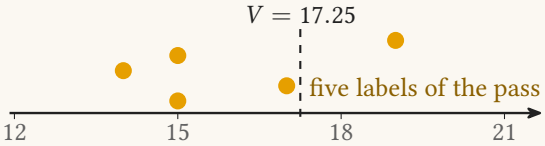
Estimating the critic is expensive



1 To label a step we wait until the day ends.

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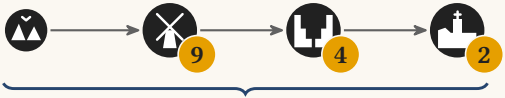


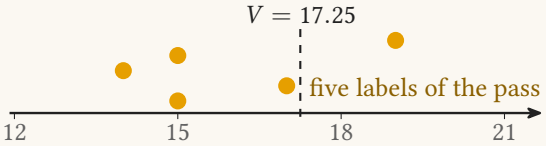


1 To label a step we wait until the day ends.

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Estimating the critic is expensive





1 To label a step we wait until the day ends.

2 A label is a sum of random coins: its spread around V is 2.82 coins.

3 The critic needs many days to average that noise out. Can we find a cheaper way?

The coins to come unfold one step at a time

$$G_t = r_t + \gamma\, r_{t+1} + \gamma^2\, r_{t+2} + \cdots + \gamma^{T-t}\, r_T$$

1 G_t adds the coins from step t to the last step T of the day, each one γ times smaller than the one before. On Mara's road $T = 3$ and $\gamma = 1$.

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&= r_t + \gamma\, \underbrace{\left(r_{t+1} + \gamma\, r_{t+2} + \cdots + \gamma^{T-t-1}\, r_T\right)}_{G_{t+1}} = r_t + \gamma\, G_{t+1}
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$$

$G_3 = r_3 = 2$ $\qquad$ $G_2 = r_2 + G_3 = 4 + 2 = 6$ $\qquad$ $G_1 = r_1 + G_2 = 9 + 6 = 15$

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3 Day 1 of Part 2, from the last step back.

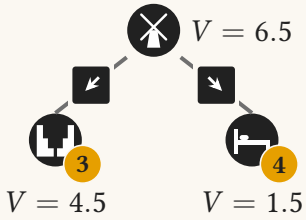
Averaging it gives the Bellman equation

$$V(s_t) = \underbrace{\mathbb{E}\big[\, r_t + \gamma\, V(s_{t+1}) \mid s_t \big]}_{\text{the average over what can follow } s_t} \;\approx\; r_t + \gamma\, V(s_{t+1})$$

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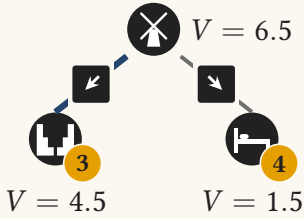


1 Average both sides of $G_t = r_t + \gamma G_{t+1}$, given that Mara stands at s_t: over the road, the next place and the luck.

2 At the market:
$\frac{1}{2}(3 + 4.5) + \frac{1}{2}(4 + 1.5) = 6.5$.

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$$V(s_t) = \underbrace{\mathbb{E}\big[\, r_t + \gamma\, V(s_{t+1}) \mid s_t \big]}_{\text{the average over what can follow } s_t} \;\approx\; \underbrace{r_t + \gamma\, V(s_{t+1})}_{\text{one sample of it}}$$



Day 1 took the first road: paid 4 coins, the usual 3 and a lucky one.

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2 At the market:
$\frac{1}{2}(3 + 4.5) + \frac{1}{2}(4 + 1.5) = 6.5$.

3 One step gives one sample: $4 + 4.5 = 8.5$.

After every step the critic can learn

$$y_t = r_t + \gamma\, V_\phi(s_{t+1})$$

the target: the table's number for the next place stands in for the rest of the day

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$$y = 4 + 1.52 = 5.52$$

$$V_\phi(\text{market}) = 4.14 + 0.2 \times (5.52 - 4.14) = 4.42$$

1 A step of size β with y_t held fixed: a semi-gradient.

2 In the algorithm the label G_t of the critic's step becomes y_t. The multiplier is still $G_t - V_\phi(s_t)$.

8

Part 8

GAE

A third multiplier uses the critic's numbers

the box Ψ_t of the gradient formula holds

Part 4 $\Psi_t = \boxed{G_t}$

the coins to come

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$$\Psi_t = \boxed{G_t}$$

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Parts 5 to 7

$$\Psi_t = \boxed{G_t - V(s_t)}$$

the coins to come minus the critic's number

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1 The box has held the coins to come, then the coins to come minus the critic's number.

2 A third option, $\hat{A}_t$, starts from the critic's numbers.

3 One number λ sets how much of the day it lets in.

Step 1: a surprise for every step, all at once

$$\delta_t = r_t + \gamma\, V(s_{t+1}) - V(s_t) \quad \text{for } t = 1, \ldots, T$$

1 The surprise of a step: what Mara got, plus what the critic expects next, minus what it expected here.

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	step 1	step 2	step 3
			
r_t	9	4	2
$+\gamma V(s_{t+1})$	6.5	4.5	0
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3 Positive: the step beat what the critic expected. Negative: it fell short.

Step 2 sweeps from the last step to the first

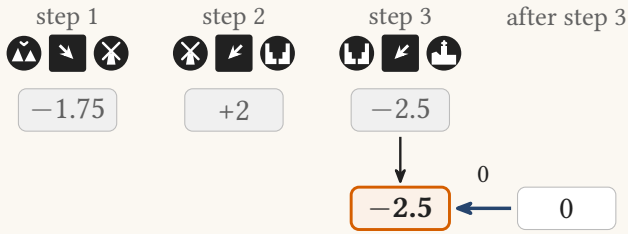
$$\hat{A}_{T+1} = 0$$
$$\hat{A}_t = \delta_t + \gamma\lambda\,\hat{A}_{t+1} \qquad \text{for } t = T, T-1, \ldots, 1$$

	step 1	step 2	step 3	after step 3
				
δ_t	-1.75	$+2$	-2.5	
$\hat{A}_t$				0

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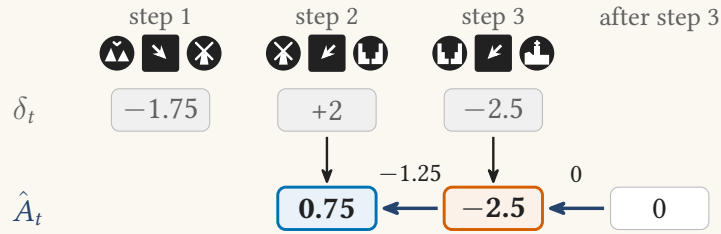


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2 Each step adds its surprise to $\gamma\lambda$ times the value of the step after it. Here $\gamma = 1$ and $\lambda = 0.5$.

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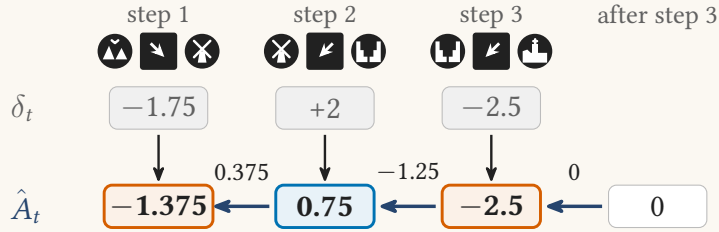


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$$\hat{A}_{T+1} = 0$$
$$\hat{A}_t = \delta_t + \gamma\lambda\,\hat{A}_{t+1} \qquad \text{for } t = T, T-1, \ldots, 1$$

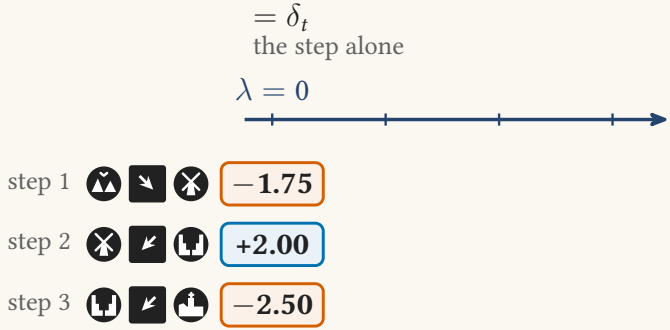


1 The sweep starts after the last step, with $\hat{A}_{T+1} = 0$.

2 Each step adds its surprise to $\gamma\lambda$ times the value of the step after it. Here $\gamma = 1$ and $\lambda = 0.5$.

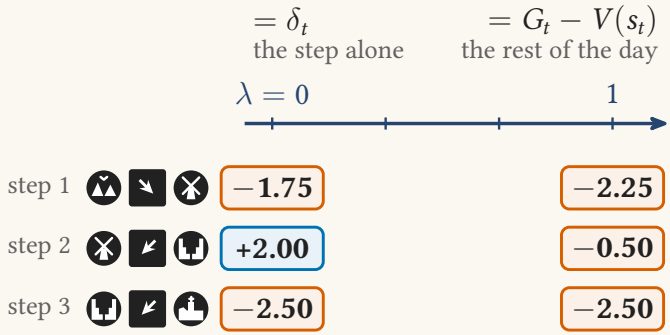
3 It runs from the last step to the first.

One number λ dials between two multipliers



1 The multiplier is $\Psi_t = \hat{A}_t$, the value the sweep ends with at step t.

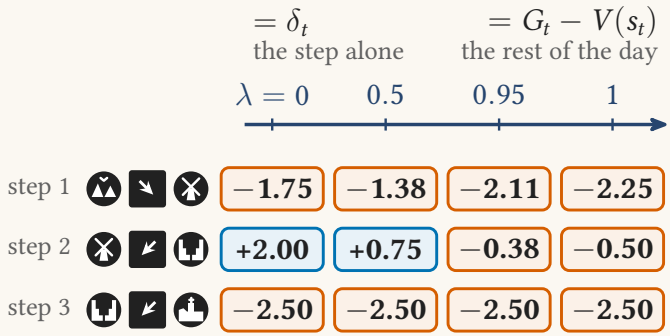
One number λ dials between two multipliers



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2 $\lambda = 0$ gives δ_t. $\lambda = 1$ gives $G_t - V(s_t)$, the multiplier of Part 5: $15 - 17.25 = -2.25$ at step 1.

One number λ dials between two multipliers



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2 $\lambda = 0$ gives δ_t. $\lambda = 1$ gives $G_t - V(s_t)$, the multiplier of Part 5: $15 - 17.25 = -2.25$ at step 1.

3 In between, λ sets how much of the rest of the day is let in.

9

Part 9

PPO

REINFORCE is gradient descent on a loss

$$L(w) \;=\; -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 This is Part 4's loss: minus the multiplier times the log chance of the road taken.

REINFORCE is gradient descent on a loss

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

$$\nabla L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \underbrace{\boxed{\Psi_t^{(n)}} \nabla_w \log \pi_w(a_t^{(n)} \mid s_t^{(n)})}_{\text{minus the estimate of Part 4}}$$

1. This is Part 4's loss: minus the multiplier times the log chance of the road taken.
2. The multiplier is a plain number: the gradient is minus the estimate of Part 4.

REINFORCE is gradient descent on a loss

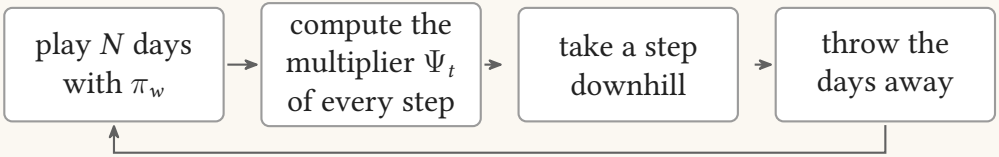
$$L(w) \;=\; -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

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minus the estimate of Part 4

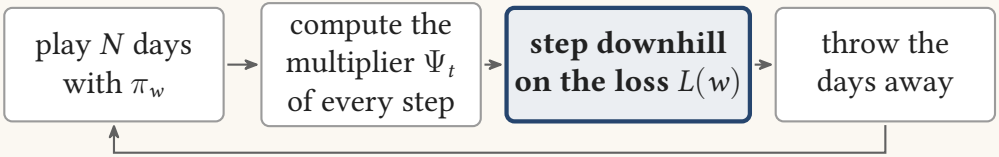
1. This is Part 4's loss: minus the multiplier times the log chance of the road taken.
2. The multiplier is a plain number: the gradient is minus the estimate of Part 4.
3. A step downhill, $w \leftarrow w - \alpha\,\nabla L(w)$, is a step uphill on $\mathcal{J}$: Part 4's update.

PPO keeps the algorithm and changes the loss



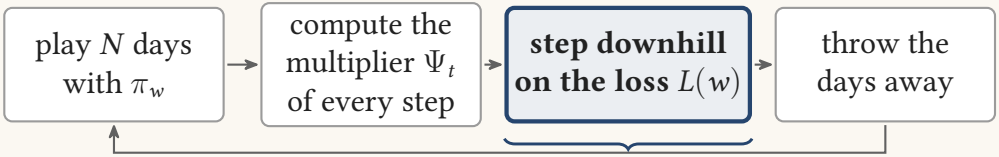
1 The algorithm of Part 4 stays: play, compute the multiplier, step, discard.

PPO keeps the algorithm and changes the loss



1. The algorithm of Part 4 stays: play, compute the multiplier, step, discard.
2. The step is gradient descent on a loss: the loss decides what the step does.

PPO keeps the algorithm and changes the loss



Part 4 $L(w) = -\frac{1}{N} \sum \sum \Psi_t \log \pi_w(a_t \mid s_t)$

Part 9 a new loss, built in the next frames

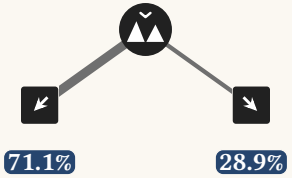
1 The algorithm of Part 4 stays: play, compute the multiplier, step, discard.

2 The step is gradient descent on a loss: the loss decides what the step does.

3 Autograd differentiates whatever loss we write. PPO writes a new one.

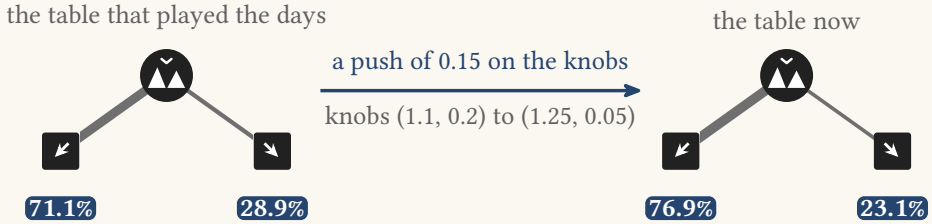
The loss remembers the table that played the days

the table that played the days



1 Each step stores the chance its road had in the table that played the day: $\pi_{\text{old}}(a_t \mid s_t)$.

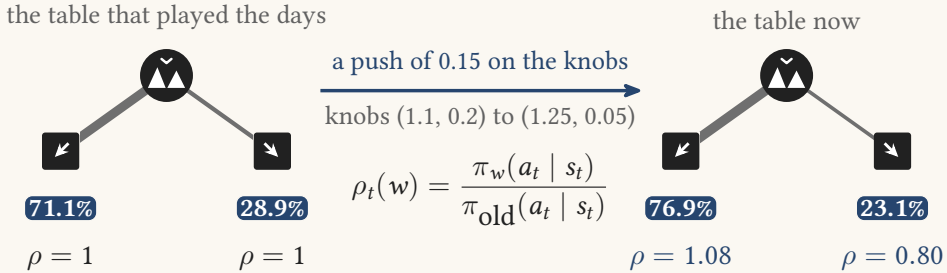
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2 After a push the table gives other chances: 76.9% and 23.1% at the pass.

The loss remembers the table that played the days



1 Each step stores the chance its road had in the table that played the day: $\pi_{\text{old}}(a_t \mid s_t)$.

2 After a push the table gives other chances: 76.9% and 23.1% at the pass.

3 The ratio is the new chance over the old one: 1.08 for ↙ and 0.80 for ↘.

The first loss multiplies the ratio by the multiplier

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

1 Part 4's loss has the log chance of the road taken in every term.

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replace it by the ratio

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \rho_t^{(n)}(w)$$

1. Part 4's loss has the log chance of the road taken in every term.
2. The new loss uses the ratio instead. At the old table every ratio is 1.

The first loss multiplies the ratio by the multiplier

$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \log \pi_w(a_t^{(n)} \mid s_t^{(n)})$$

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$$L(w) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{t=1}^{T} \boxed{\Psi_t^{(n)}} \quad \rho_t^{(n)}(w)$$

1. Part 4's loss has the log chance of the road taken in every term.
2. The new loss uses the ratio instead. At the old table every ratio is 1.
3. So the gradient is Part 4's and the first step is Part 4's step.